

IngRICE: A Unified Bioinformatics Portal for Rice Research

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I. INTRODUCTION

A. Background of the Study

Rice is an important crop that feeds billions of people around the world, specifically in Asia and part of the Pacific. It is a key part of global food security and economic stability [1]. Because of its wide genetic variety and ability to grow in different environments, rice has been the focus of many research studies to solve problems like increasing food demand, climate change, and changing consumer needs [2]. These problems have led to the development of new technologies that improve our understanding of rice genetics and help produce better crop varieties.

The sequencing of the rice genome was a big breakthrough in plant science. Researchers have advanced rice genomics at a high speed in the last century. In 1998, the Rice Genome Research Program spearheaded the uncovering of the complete genomic sequence for rice species [3]. In recent years, “The 3000 (3K) Rice Genome Project” has significantly contributed in exposing genomic diversity from 3000 accessions in sources of rice germplasms around the world [4]. It then posed a major challenge of comprehensively and systematically mining this giga-dataset of publically available genome sequences (averaging 14× depth of coverage) to derive genotypic variation to functional variation with the ultimate goal of creating new and sustainable rice varieties that can support a future world population that will approach 9 billion by 2040 [4].

While these advances have provided unprecedented access to rice genomic data, they have also led to generation of the tremendous amount of data, development of species-specific [5] databases, and increasing number of Bioinformatics tools [6], [7], [8] designed to analyze and interpret these datasets [9]. However, these tools are often dispersed across different platforms, repositories, and research institutions. This fragmentation makes it difficult and time-consuming for researchers to locate, access, and utilize these resources effectively. This yields to a growing demand for a unified Bioinformatics portal — a one-stop-shop that consolidates rice genomic datasets and related tools into a single accessible platform. Such a portal would streamline data access, improve interoperability between tools, and enhance collaborative research efforts in the rice genomics community.

Tripal, a software framework based on the Drupal platform, is a good option for managing rice genetic data. It is already used in many plant genomics projects, such as the BeeBiome [10] and ICGRC [11] portals. Tripal can handle large and complex datasets and present them in ways that are easy for users to work with. Using Drupal and Tripal, a rice-focused

Bioinformatics tool can bring together important data in one place, making it easier for researchers to study and share their findings.

This study will feature a bioinformatics portal for rice research. This portal will combine modern genomic technologies with user-friendly features to help researchers analyze rice data and work together more effectively. By addressing the tedious process of locating rice bioinformatics tools developed using the massive datasets posed by the 3K Rice Genome Project and getting access to these tools through the traditional and manual authentication process, the portal can help rice scientists and researchers automate these problems to develop solutions for improving rice production and securing the global food supply.

B. Statement of the Problem

Advances in rice genomics sparked by the efforts of “The 3000 Rice Genome Project” have produced an unprecedented volume of genomic data, along with a growing number of species-specific databases and bioinformatics tools for data analysis and interpretation. However, these resources are often fragmented, and dispersed across various platforms, repositories, and research institutions. This fragmentation poses significant challenges for researchers, making it difficult and time-consuming to locate, access, and integrate these tools and datasets.

The lack of a unified and centralized portal for rice genomic resources hinders interoperability between tools and complicates collaborative research efforts. Thus, a demand for a comprehensive bioinformatics portal — a one-stop platform that consolidates rice genomic datasets, databases, and tools. Such a portal would streamline access, improve efficiency, and foster enhanced collaboration within the rice research community.

C. Significance of the Study

This study aims to contribute to the field of Bioinformatics, particularly for rice research. This will address the need for an accessible, specialized, and scalable Bioinformatics portal. By utilizing Drupal and Tripal 4 technologies, this research aims to provide a centralized, user-friendly system for rice biological data sharing, visualization, and collaborative research among rice researchers globally. This portal will specifically aim to facilitate the integration of multi-omics data (genomic, transcriptomic, proteomic) into a single interface in the near future, catalyzing the advancement of rice breeding programs. The outcomes are expected to benefit rice researchers, breeders, and geneticists, supporting the sustainable improvement of rice varieties crucial to global food security.

D. Scope and Limitations

This research focused on developing a Bioinformatics portal for rice breeding and genetics research, integrating rice datasets and bioinformatics tools. The portal supports data visualization, pathway mapping, and secure, personalized data access designed for rice breeders, geneticists, bioinformatics researchers, and other stakeholders. It offers role-based access and conditional tool/feature rendering, allowing customized usability. The portal aims for sustainability by implementing a simple yet scalable architecture, making it adaptable for future genomic tools and evolving research needs.

The portal aims to support a wide range of rice datasets and bioinformatics tools but initially focuses on loading published datasets for *Oryza sativa* Japonica group, open-source visualization tools, and developed in-house tools by IRRI (International Rice Research Institute). Extending functionality to other data types or research tools may require additional development. While user-friendly, some training may be necessary for users with limited bioinformatics knowledge. The system's performance and scalability depend on the hosting infrastructure, which may pose challenges for institutions with limited computational resources. The selection of participants for the project's testing and evaluation will be determined using convenience sampling. Despite these limitations, the portal offers an accessible and extensible foundation for enhancing rice research and breeding efforts.

II. RESEARCH OBJECTIVES

To design and develop a bioinformatics portal for rice research capable of integrating and visualizing multi-omics data. Specifically, this research aims to fulfill the following objectives:

- 1) To design and implement an efficient backend database design using TriPal and PostgreSQL following the Chado biological database schema.
- 2) To implement a front-end portal interface using the existing Drupal custom themes.
- 3) To integrate visualization tools built from rice multi-omics data, specifically Genotype Viewer, Jbrowse, Synteny Viewer, and Biological Pathways Mapping.
- 4) To develop a secure authentication-based conditional rendering for personalized data access and feature visibility, and improve interoperability between tools.

III. REVIEW OF RELATED LITERATURE

A. Impacts of Rice Research

Rice research is essential for sustaining food security and meeting consumer demands, as illustrated by recent trends in Tanzania stated in the review article. Since the 1960s, rice consumption in Tanzania has risen sharply due to urbanization and changing dietary preferences. While imported rice is often less expensive, Tanzanians favor traditional local varieties for their unique flavor and aroma, which elevates market prices [12].

Efforts to introduce foreign high-yield varieties have faced obstacles, as many lack the grain quality desired by con-

sumers. Consequently, a market-oriented research approach is recommended, emphasizing the genetic strengths of traditional cultivars while improving yield potential. Additionally, Tanzania's varied agroecological zones require targeted agronomic practices for optimizing production in specific regions [12].

The review highlights how consumer-centered rice research, aligned with local cultural and market demands, can boost food security and economic growth. Considering varied agronomic practices for specific regions, researchers must also come from diverse locations and backgrounds, thus demanding accessibility to diverse genomic information and research tools.

B. Progression of Rice Genomics Research and the Research-worthy Features of Rice

Rice has become a prominent focus in plant genomics due to its unique biological properties and agricultural significance. Over the past two decades, rice genomics research has made remarkable progress, positioning rice as a model organism for genetic studies. As detailed in 20 Years of Rice Genomics Research, scientists have sequenced the rice genome and identified genes linked to vital traits such as yield, stress resistance, and nutritional content. These insights have not only improved rice varieties but have also contributed broadly to plant science, influencing research on other crops with similar genetic characteristics [13].

Rice is considered to be a perfect model plant for plant genomics research because of its relatively small genome size (400 Mb), well-established and highly efficient transformation system, extensive genetic resources, and synteny with the sequences of other cereal crops [14]. Due to its nature of having the smallest genomic size among other major cereals, rice made a breakthrough as the first organism that had its whole genome sequenced [15]. This, combined with rice's diversity and adaptability, makes it a valuable model for understanding plant genetics and developing agricultural innovations. Recent research has leveraged gene editing and advanced biotechnologies to introduce resilience traits, such as drought and pest resistance, into rice varieties [16]. Furthermore, wild rice varieties have become a rich source of genetic material, offering traits like disease resistance and environmental adaptability that modern cultivars sometimes lack [17]. These genomic advancements highlight rice's role not only in agriculture but also as a keystone species in the study of plant genetics and evolution.

C. Existing Rice Research and Discovery

Nutritional research explores genetic variations in cultivated rice that affect qualities such as amylose content, which influences grain texture and suitability for various culinary uses. The study by Hebishy et al. examines three rice varieties with varying amylose levels, finding that higher amylose content contributes to firmer and less sticky rice, while lower levels yield softer, stickier grains. This characteristic makes rice with lower amylose, such as the Japanese cultivar Koshihikari, ideal for sushi preparation due to its desired chewiness and

shape retention when cooled [18]. Variations in amylose and amylopectin composition significantly influence sensory properties, pasting behaviors, and storage quality of rice, highlighting the importance of genetic selection in enhancing both nutritional value and culinary versatility in rice breeding.

D. Technological Advancements Furthering Rice Research

The past decade has seen rapid advancements in sequencing and genomic technologies, transforming rice research and breeding practices. Technologies like next-generation sequencing (NGS), pan-genomics, and CRISPR-Cas9 gene editing have not only increased the efficiency and accuracy of rice breeding but have also allowed scientists to move beyond traditional breeding methods to more precise, genome-assisted approaches [15]. These tools have made it possible to introduce beneficial traits, such as improved yield and stress resilience, into rice varieties at a significantly faster rate, addressing worldly challenges posed by population growth and climate change.

The shift from field-based selection to genomic-assisted breeding allows researchers to streamline the breeding process by targeting specific genes associated with desirable traits, thus reducing time and resource requirements [15]. Moreover, as sequencing and Bioinformatics tools continue to evolve, rice breeding is expected to further integrate these advanced methods, leading to faster, more targeted crop improvement. This technology-driven shift depicts the vital role of genomic tools and constructing a unified portal that gathers them in advancing rice research and accelerating the development of resilient, high-yield rice varieties suited for diverse environments.

E. Advantages of Tripal vs. widely-adopted systems for genomic and genetic web databases/portals

A study by Mansueto et al. defines Tripal as a powerful platform for managing genomic data, built on the Drupal framework and employing the Chado schema to organize biological entities and their relationships. Unlike other popular systems like InterMine and EnsemblPlant, Tripal's modular design allows for flexible data loading, visualization, and analysis through plug-in 'omics modules. Widely used in plant genomics projects, including the SouthGreen Bioinformatics platform [19], Tripal is also GMOD (Generic Model Organism Database) standards-compliant and supports collaborative content management, making it an ideal choice for community-driven research [11]. Tripal's modular design allows it to meet the distinct needs of rice researchers by supporting functionalities like secure data access, personalized feature rendering, and multi-omics data visualization. Drupal's robust content management infrastructure, combined with Tripal's Bioinformatics capabilities, ensures that a rice genomics portal can evolve alongside technological advancements, addressing both usability and security in accessing sensitive genetic data.

F. Existing Tripal Applications

In the development of Bioinformatics portals, Tripal, built on the Drupal content management system, offers a flexible framework for managing and sharing genomic data in collaborative environments. Both the BeeBiome [10] and ICGRC [11] projects illustrate how Tripal and Drupal can be combined to support multi-omics data integration, user collaboration, and dynamic content management for specific research communities.

In the BeeBiome project, Tripal's integration with Drupal facilitated the creation of a portal that manages extensive microbiome metadata across different bee species and geographical locations. The portal uses Drupal's core features, such as customizable content types and user roles, to manage access and display relevant data to specific users. Leveraging Tripal's modules, the portal integrates with Chado to organize complex datasets, such as sequence and phenotypic data, while offering analysis tools through linked applications like Galaxy, Apollo, and JBrowse. Tripal's modularity also enabled the BeeBiome portal to incorporate both public and restricted datasets, allowing researchers to perform specific analyses and access tailored information through authenticated logins. The Tripal BeeBiome portal also allows for high portability with the Docker image and can be hosted easily in the cloud or on a traditional server, without the need to install any other dependencies [10].

Similarly, the ICGRC portal, designed to support the Cannabis genomics research community, uses Drupal and Tripal to provide a collaborative platform for hosting and analyzing genomic, phenotypic, and metabolic data. The portal includes functionality to upload and visualize multi-omics datasets, including gene annotations, QTLs (Quantitative Trait Loci), and genetic markers. The ICGRC project uses Tripal's API to support automated data sharing and data retrieval, allowing researchers to seamlessly perform multi-omics analyses and collaborate on data annotations. This integration enables the portal to accommodate large datasets efficiently, with Drupal providing content management and user interface customization while Tripal supports omics-specific data modules that allow users to conduct in-depth, multi-layered analyses [11].

These studies demonstrate how the integration of Drupal and Tripal creates a comprehensive, scalable Bioinformatics portal. Drupal provides the infrastructure for user roles, authentication, and custom content management, while Tripal builds on this foundation to manage and analyze complex Bioinformatics data, making it an ideal solution for species-specific research communities.

IV. METHODOLOGY

The first part of the methodology provides an overview and preliminaries, including the hardware and machine specifications that will be used in this study. The succeeding subsections will address the specific objectives of this study. The back-end architecture and technologies, along with the themes utilized in building the front end will be defined. Data sources and processing for the integration of bioinformatics tools will also be discussed side-by-side with reference to the

ICGRC web portal development workflow. Lastly, the metrics and evaluation methods concerning the research objectives will also be constituted in this section.

A. Machine Specifications

The machine used in developing the model and application is a Lenovo ThinkPad X1 Carbon Gen 8 with Microsoft Windows 11 Pro as the operating system. A WSL (Windows Subsystem for Linux) was installed on the machine. This environment handled the compilation, hosting, and running of the program. The system consisted of an Intel(R) Core(TM) i5-10310U CPU 1.70GHz, 2208 Mhz, 4 Core(s), 8 Logical Processor(s), and a 16-gigabyte (GB) Random Access Memory (RAM).

B. System Specifications and Implementations

The Bioinformatics portal was designed using a modular and scalable architecture to integrate diverse datasets and tools for rice research. The portal is built on Drupal, a PHP-written CMS (Content Management System/Framework), with implemented Tripal extension, a web-based platform designed for biological databases. Here is the architectural specifications of the project.

a) Technology Stack:

- **Framework:** Drupal 10.3.x with the Tripal 4 extension.
- **Database:** PostgreSQL, optimized for managing Chado schema biological data.
- **Server:** Apache2 with PHP-FPM for handling web requests.
- **Containerization:** Docker for consistent deployment environments.

b) Development Environment:

A local development environment was set up using Docker, with containers for PHP-Apache, PostgreSQL, and custom tools.

c) Database Setup:

The PostgreSQL database was configured with the Chado schema to store biological data. Biological data for *Oryza sativa* (rice) was curated using Tripal tools.

d) Frontend Development:

A custom theme based on Radix; an original component-based theme with Bootstrap 5, Sass, and other built-in supports, was implemented for intuitive user experience and cross-device compatibility.

e) Deployment:

The finalized portal was dockerized; for continuous integration and delivery, and deployed on a cloud platform (AWS EC2 instance).

C. Data Integration

The portal integrated datasets related to rice varieties and other organisms, enabling researchers to query, analyze, and visualize complex biological relationships.

a) Data Sources:

- Genomic sequences and annotations retrieved from publicly available repositories (e.g., NCBI, NCBI-GEO, IRR1).
- Gene expression datasets that were specific to *Oryza sativa* (rice) varieties.
- Reciprocal best hit (RBH) gene mappings for comparative genomics analysis with other species.

b) Data Processing:

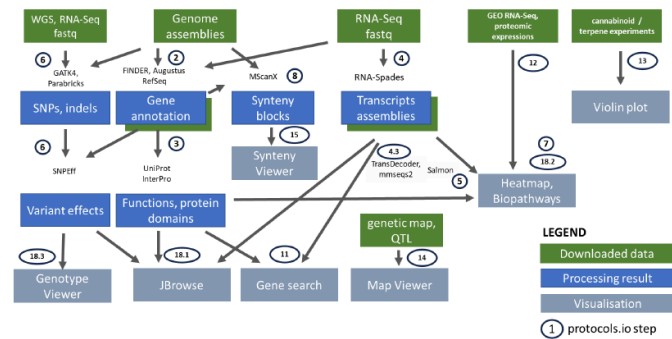


Fig. 1: Detailed program development workflow for file loading, processing, and visualization from the ICGRC web portal [11]

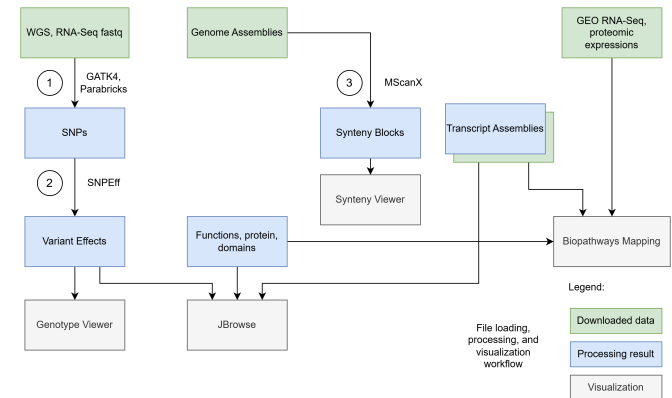


Fig. 2: Visualization of the subset development workflow to be implemented in this study

Figure 2 illustrated the comprehensive workflow designed for loading, processing, and visualizing multi-omics data in the bioinformatics portal. This workflow builds upon the foundational framework proposed in the ICGRC bioinformatics platform paper (refer to Figure 1). Loaded data were modified to include rice biological datasets while maintaining the structure of the workflow. The green boxes represent downloaded data sources to be loaded, the blue boxes represent processing

results given the processing tools used, and the white boxes represent the interactive visualization tools to be integrated into the website.

D. Portal Tools and Features

These are the bioinformatics tools loaded with public biological data sources that will be integrated into the portal. All of these tools were referenced from the ICGRC web portal [11].

- Synteny Search - Tripal SynView will be utilized to showcase synteny blocks, highlighting genomic conservation across rice varieties and other species.
- Biological Pathways Mapping - The non-Tripal module was developed to visualize metabolic pathways and gene ontology terms using MapManJS. The Iframe Drupal module was customized for seamless integration.
- Genome Browser (JBrowse) - Built using JBrowse, this tool will allow the display of various sources of genomic data with its corresponding reference sequences and scaffolds.
- SNP-Seek Genotype Viewer - This tool will query the SNP genotype matrix constrained by genomic region and genotype dataset. The region is defined by the reference assembly, chromosome/contig, start and end or by selecting the gene locus. The interface uses the SNP-Seek software and database infrastructure developed by (Mansueto 2017) for the International Rice Informatics Consortium [11].

their classification either as a Tripal or a non-Tripal tools and dataset prerequisites. Integration of Tripal modules follow the traditional pipeline of utilizing Tripal built-in features, while an external environment in a separate container was created and allocated to non-Tripal modules for modularized debugging. External modules will be embedded into the portal using the iframe html tag.

E. Validation and Testing

a) Usability Testing:

Program usability will be evaluated by a usability testing session. The participants will include both experienced bioinformatics users and novice researchers to ensure a diverse range of feedback. Each participant will perform predefined tasks, such as navigating to specific datasets, visualizing genomics data, and others. The metrics used were the percentage of personally completed tasks, the average time spent completing each task, and a System Usability Scale (SUS).

b) Functionality and Security Testing:

This testing will focus on verifying core functionalities, such as the accurate visualization of datasets and the clarity and interpretability of generated outputs. Additionally, the system's role-based tool/feature visibility and secure authentication mechanisms will be assessed to ensure data protection. Manual testing will be used considering the portal's functionality, data accuracy, and level of security based on a pass/fail system for binary classifications and a 5-point Likert scale for rating-based criteria from a score sheet.

c) Assessing System Architecture Compatibility and Ease of Development:

This evaluation aims to determine how effectively the Bioinformatics portal's architecture supports the seamless integration of datasets and tools while ensuring the system can be easily handed over to bioinformatics experts with limited technical background. The assessment involves bioinformatics experts and researchers performing guided tasks such as updating themes, configuring modules, importing new datasets, integrating other tools, and configuring pathways and mappings. The ease of completing these tasks will be measured by tracking the success rate, the average time taken, and self-reported ease-of-use ratings on a 5-point Likert scale. Feedback on potential challenges or gaps during the handover process will be noted to ensure the system can be effectively maintained and extended by non-technical users.

System engineers or IT professionals familiar with web-based bioinformatics tools will review the system documentation and architecture. They will provide feedback on code maintainability, modularity, and scalability. Their insights will be evaluated through qualitative feedback and ratings on a 5-point Likert scale for aspects like documentation clarity, scalability, and system robustness.

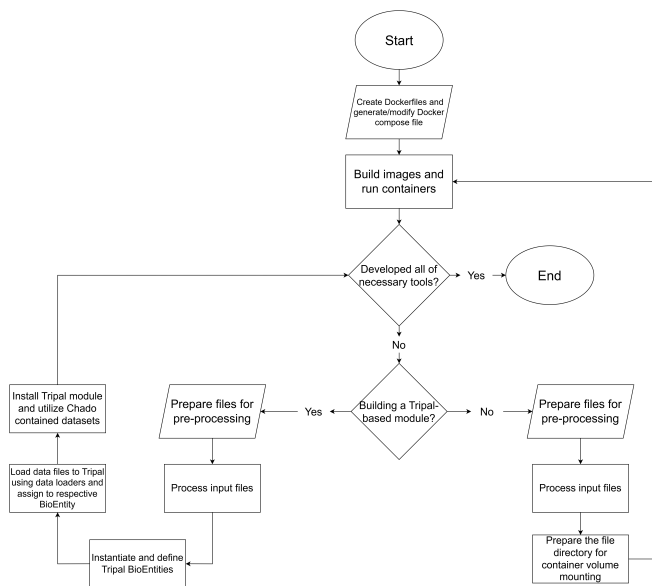


Fig. 3: Proposed program development workflow for tool integration and containerization

Illustrated in Figure 3 is the proposed modularized development pipeline for the study. Development started with initializing and building a bare Drupal website inside a Docker container. Modules and tools were integrated according to

V. APPENDIX

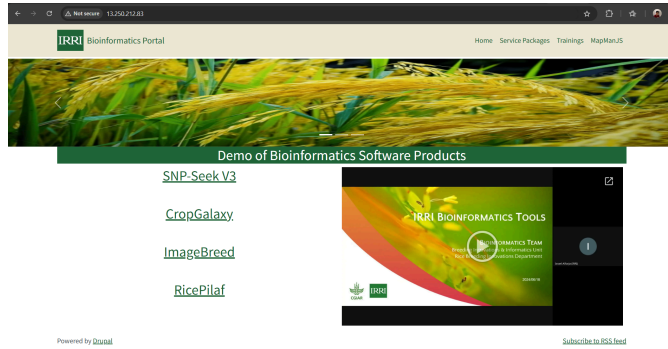


Fig. 4: Wireframe of landing page for anonymous users

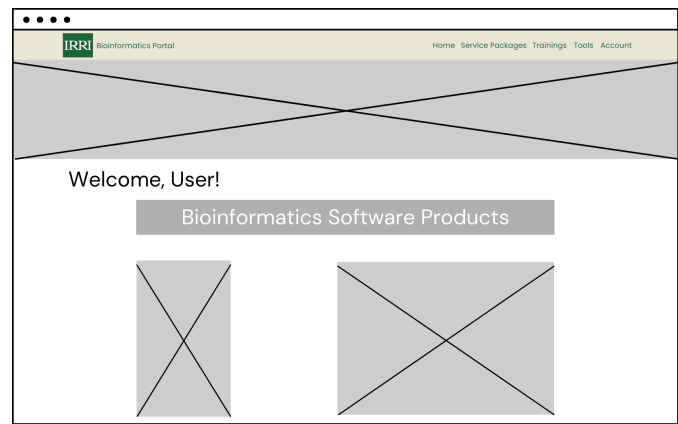


Fig. 7: Wireframe of landing page for authenticated users

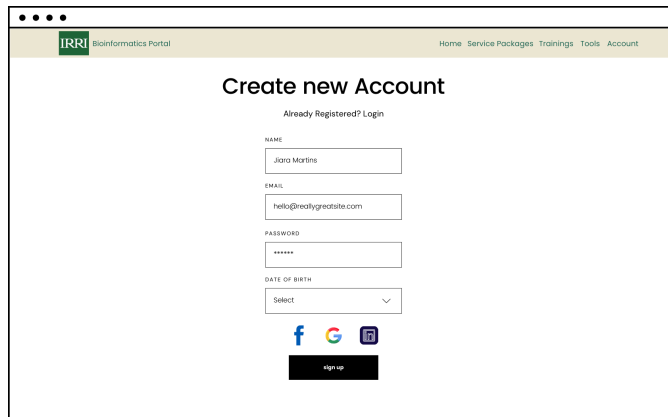


Fig. 5: Wireframe of sign up page

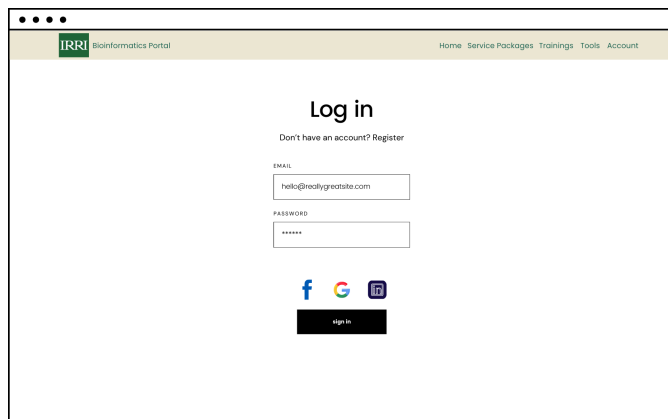


Fig. 6: Wireframe of login page

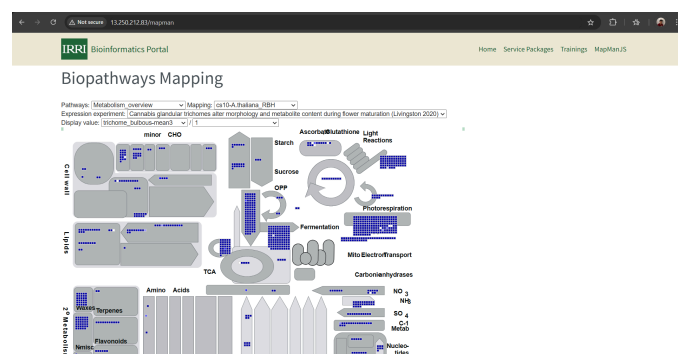


Fig. 8: Wireframe of custom tools

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